Network Troubleshooting, Management, Monitoring, and Optimization

1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.



```
C:\Users\Prachi Patel>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-QBBAHUGO
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-12
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::71a6:fde2:3cca:f593%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 822738983
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2C-6E-CB-B2-F8-9E-94-65-B8-B9
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : F8-9E-94-65-B8-BA
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : FA-9E-94-65-B8-B9
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
   Physical Address. . . . . . . . . : F8-9E-94-65-B8-B9
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2401:4900:8fc7:ac23:384d:1ad0:3664:ab48(Preferred)
   Temporary IPv6 Address. . . . . . : 2401:4900:8fc7:ac23:c152:913f:2cf4:d59f(Preferred)
   Link-local IPv6 Address . . . . . : fe80::917e:2d0a:7f66:9a30%17(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.13(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, October 8, 2026 9:01:09 PM
   Lease Expires . . . . . . . . . . : Friday, October 9, 2026 9:01:10 PM
   Default Gateway . . . . . . . . . : fe80::e88:32ff:fe9f:e60c%17
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 184065684
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2C-6E-CB-B2-F8-9E-94-65-B8-B9
   DNS Servers . . . . . . . . . . . : fe80::e88:32ff:fe9f:e60c%17
                                       192.168.1.1
                                       fe80::e88:32ff:fe9f:e60c%17
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : F8-9E-94-65-B8-BD
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\Prachi Patel>
```

IPv4 Address: 192.168.56.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Servers: 192.168.1.1

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

I first disconnected my Wi-Fi and ran the following command in Command Prompt:

```
C:\Users\Prachi Patel>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\Prachi Patel>
```

Before reconnecting: The ping was unsuccessful and showed messages such as “Request timed out” or “General failure”, indicating that my computer could not reach Google because there was no active network connection.

Then, I reconnected the Wi-Fi and ran the same command again:

```
C:\Users\Prachi Patel>ping www.google.com

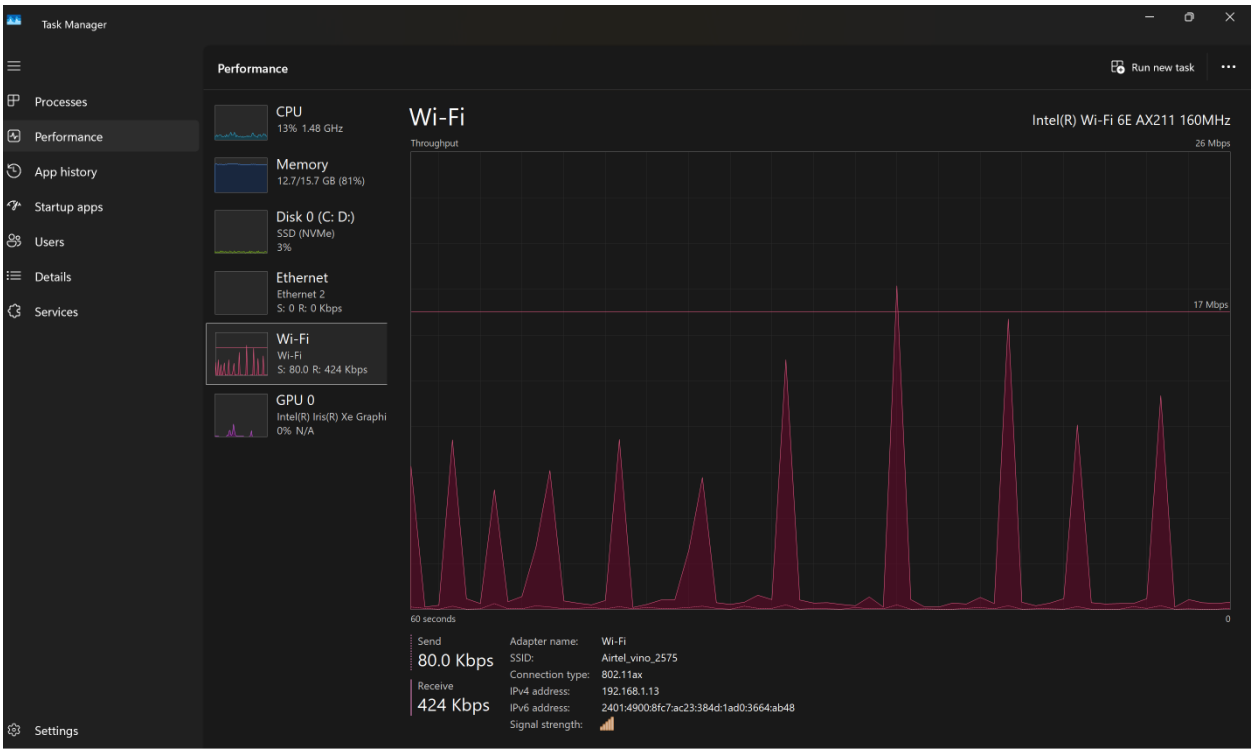
Pinging www.google.com [2001:4860:482d:7700::] with 32 bytes of data:
Reply from 2001:4860:482d:7700::: time=44ms
Reply from 2001:4860:482d:7700::: time=39ms
Reply from 2001:4860:482d:7700::: time=96ms
Reply from 2001:4860:482d:7700::: time=139ms

Ping statistics for 2001:4860:482d:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 39ms, Maximum = 139ms, Average = 79ms

C:\Users\Prachi Patel>
```

After reconnecting: The ping was successful and displayed “Reply from …” along with the time in milliseconds (ms). This means my computer had successfully connected to the network and could communicate with Google's server.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.



**4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.**

I used Wireshark to capture network traffic for approximately 30 seconds while browsing a website such as Flipkart. After capturing the packets, I used the filter bar and entered:

http or https

This displayed packets related to HTTP/HTTPS traffic.

3 Types of Information Visible

1. Source and Destination IP Address – Shows the IP addresses of the device sending and receiving the packets.
2. Protocol Information – Shows whether the packet uses HTTP, HTTPS, TCP, TLS, etc.
3. Packet Size and Timing – Shows the length/size of each packet and the time at which it was captured.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

I used ChatGPT to generate the following troubleshooting checklist for improving home network speed and reliability while gaming:

Troubleshooting Checklist:

1. Check internet speed: Run a speed test to check download speed, upload speed, and ping/latency.
2. Restart the router: Turn off the Wi-Fi router, wait for about 30 seconds, and turn it back on to refresh the connection.
3. Use Ethernet cable: Connect the gaming device directly to the router using an Ethernet cable for a more stable and lower-latency connection.
4. Reduce background traffic: Close downloads, streaming services, cloud backups, and other applications using the internet.

5. Move closer to the router: If using Wi-Fi, keep the gaming device closer to the router and avoid walls or other obstacles.
6. Use the 5 GHz Wi-Fi band: Connect to 5 GHz Wi-Fi when available because it generally provides faster speeds and less interference than 2.4 GHz.
7. Update router firmware: Check for firmware updates to improve the router's performance, security, and reliability.